# MATHEW M. POTTS


RESEARCH SCIENTIST

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## EDUCATION

Doctorate of Philosophy, *Physics* May 2021 – May 2022

University of Utah, Salt Lake City, Utah

## SKILLS

*Programming languages:* Python, Java, C, C++, R, CERN ROOT, Geant4, MATLAB, Mathematica, LabVIEW

*Markup Languages:* HTML, CSS, LaTeX

*Linkedin Skill Badges:* Angular, React.js, Google Cloud Platform, Java, Microsoft Azure, Amazon Web Services, Machine Learning, Python

*Miscellaneous:* Strong analytical problem-solving experience, excellent verbal and written communication skills, and great in collaborative environments.

## RESEARCH EXPERIENCE

National Lab Research May 2023 – Present

Department of Physics, Pacific Northwest National Laboratory

Post Doctorate Research Assistant

- Dilution fridge and clean lab experience when working with SuperCDMS or BeEST
- National Security work; Q clearance
- Generation and analysis of Geant4 and CORSIKA simulations for various projects

Neutrino Research May 2022 – Oct. 2023

Trinity Neutrino Observatory and EUSO-SPB2

Department of Physics, Georgia Institute of Technology

Postdoctoral Research Fellow

- Designed the remote observation procedures, software, and hardware for the Trinity Demonstrator telescope.
- Built and tested all aspects of the Trinity Demonstrator telescope.
- Designed and tested the cooling system for the Cherenkov telescope of the EUSO-SPB2 experiment.

- Performed simulation using CORSIKA, GrOptics, and C++ in order to analyze the performance of the full Trinity Observatory and Demonstrator telescope.

Cosmic Rays Research Aug. 2015 – Sept. 2023

The Telescope Array (TA) Cosmic Ray Observatory

Department of Physics and Astronomy, University of Utah

Research Advisor: Charles Jui

Thesis: Ultra-high Energy Cosmic Ray Energy Spectrum using Hybrid Analysis with TAx4

- Analyzed the data for the new TA expansion, TAx4, to calculate the energy spectrum of ultra-high energy cosmic rays using hybrid detection.
- Maintained and operated surface and fluorescence detectors for cosmic ray data collection.
- Linux systems administrator of the Telescope Array's data server and computational cluster at the University of Utah.
- Analyzed fluorescence detector's sensitivity to energy and development of cosmic ray interactions in the atmosphere through Monte Carlo simulation. The reconstructed information was analyzed by binning parameters into histograms and fitted using CERN's ROOT data analysis framework.

Accelerator Research Aug. 2015 – May 2023

sFLASH Collaboration

Stanford Linear Accelerator Center, National Accelerator Laboratory 0

- Performed two experiments in 2016 and 2018 aimed at measuring the air fluorescence yield at Stanford Linear Accelerator Center. The air fluorescence yield is a important quantity in determining how much light is produced in cosmic ray air showers.
- Performed photo-multiplier tube calibration with a UV LED diode.
- I assisted in taking the data when the beam was running and was in charge of keeping an experimental log of each data run.
- Analyzed beam run data to find 'golden runs', runs where the beam energy was stable.

Astronomy Research Aug. 2014 – May 2015

Salt Lake Community College, Salt Lake City, Utah

Research Advisor: Jonathan Barnes

Senior Project: A Closer Look at the KOI-22 Light Curve

- Examined the Kepler exoplanet data to see if we could find evidence of precession in the light curves of "Hot Jupiter" systems that could hint at additional exoplanets.

- Found a trend in the difference of the orbital times of KOI-22 which repeated every sixty-two orbits. This may suggest that a body is perturbing its orbit slightly.